

10 (**long**) The matrix $\mathbf{A}$ is given by

$$\mathbf{A} = \begin{bmatrix} 1 & a & 0 \\ a & 1 & 0 \\ 0 & 0 & a \end{bmatrix}$$

where a is a positive constant.

(a) Find the eigenvalues and eigenvectors of $\mathbf{A}$. [10]

(b) The matrix exponential is defined as

$$e^{\mathbf{A}} = \mathbf{I} + \mathbf{A} + \frac{\mathbf{A}^2}{2!} + \frac{\mathbf{A}^3}{3!} + \dots = \sum_{n=0}^{\infty} \frac{\mathbf{A}^n}{n!}$$

The matrix $\mathbf{A}$ can be expressed as $\mathbf{A} = \mathbf{U}\mathbf{\Lambda}\mathbf{U}^{-1}$, where $\mathbf{U}$ is a proper orthogonal matrix and $\mathbf{\Lambda}$ is a diagonal matrix.

(i) Find the matrix $\mathbf{A}^n$ for an arbitrary non-negative integer n . [8]

(ii) Find the matrix $e^{\mathbf{A}}$, giving your answer as an explicit expression in terms of e and a . [9]

(iii) If a is small, find an approximation to $e^{\mathbf{A}}$. [3]